

User Manual:

1. Add Tickets
 - a. Open add ticket form by typing In terminal “python -m app.main”
 - b. Enter Student number(must start with “ST”), student name, category, priority, description and status if needed
 - c. Click save ticket
 - d. Success message will display in green
2. View, Search, Filter Tickets
 - a. Open view tickets by typing “python -m app.view_ticket”
 - b. Use the search box to search for a ticket ID or student ID
 - c. Use status/priority filters or sort by date/priority
3. Update/Archive ticket
 - a. Click on the ticket and click on the drop down to change the status or priority
 - b. Click Update Ticket and the changes will be saved and logged in a new table and the new status will be saved in closing comment.
 - c. Click on a ticket and click on archive ticket and a confirmation message will be displayed.
4. Dashboard/Report
 - a. In terminal type in “python -m app.reports”
 - b. The dashboard will display the counts of each ticket
 - c. Click on export csv to show the data in a csv

Technical Report:

1. System Overview:
 - a. Built with Python, Tkinter GUI, SQLite database.
 - b. Modular files: main.py (add), view_tickets.py (view/update/archive), reports.py (dashboard/report), database.py (schema).
2. Database Schema:
 - a. tickets table: ticket_id, student_number, student_name, category, description, priority, status, date_logged, closing_comment.
 - b. change_log table: log_id, ticket_id, change_type, old_value, new_value, date_changed.
3. Security Features:
 - a. Validation of required fields.

- b. Rejects unsafe <script> input.
 - c. Reports exclude personal identifiers.
 - d. Clear error messages.
- 4. Testing Summary:
 - a. 12 test cases created (normal, missing, invalid data).
 - b. Bugs logged and fixed (typos, missing labels, wrong export path).
 - c. All retested successfully.

Maintenance Notes:

- Run `python -m app.database` to initialize tables.
- Database file: `data/tickets.db`.
- Report export: `data/report.csv`.
- Backup `data/` folder regularly.

Final demonstration checklist:

- Add Tickets
- View/Search/Filter tickets
- Update/Archive tickets
- Reports shows counts of tickets
- Validation rules in place

Learner reflection:

I learnt the importance of preparation and design before coding.

I learnt how to validate form and ensure security.

Learnt how to work with Tkinter and SQLite